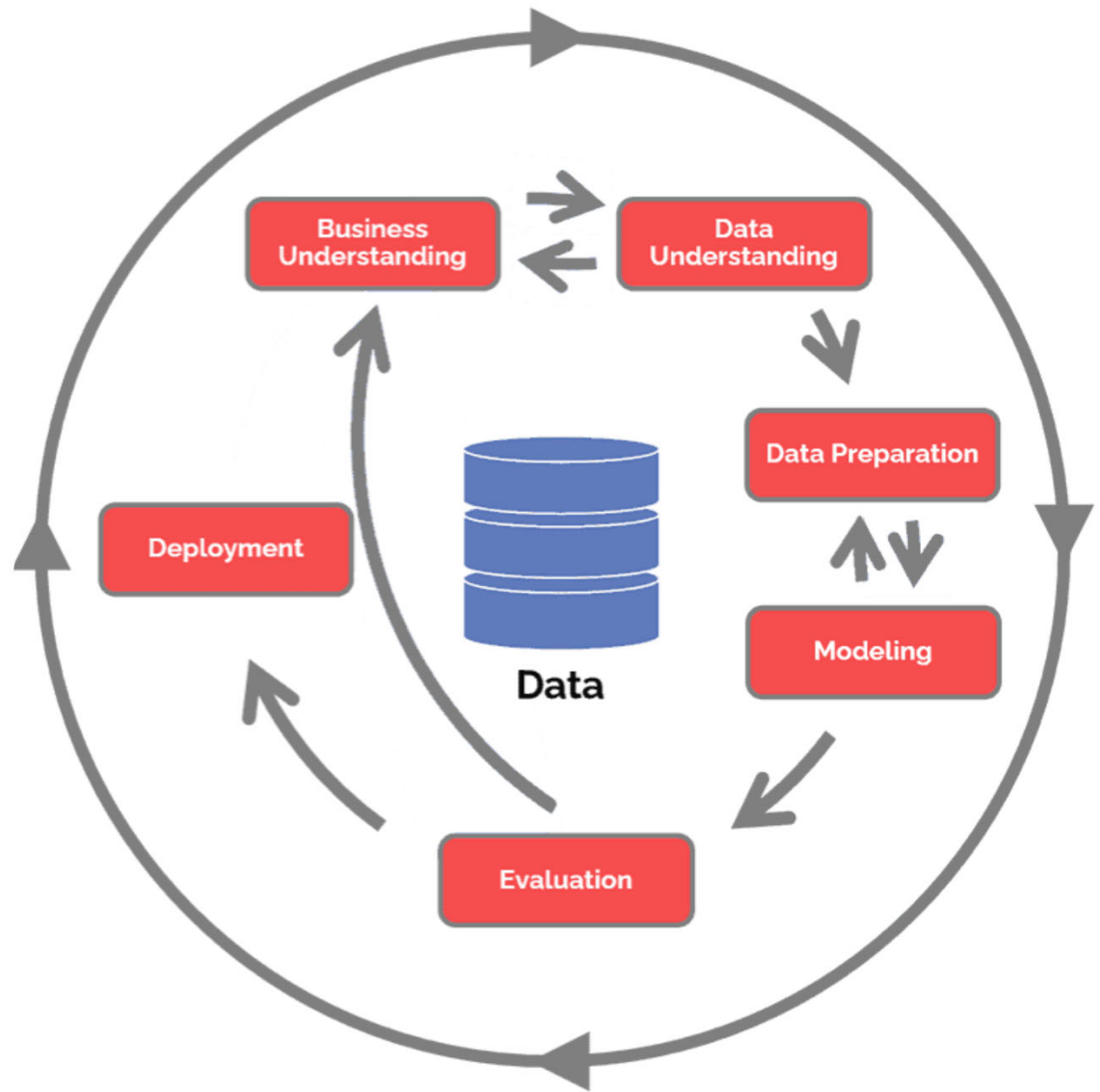


Organizational Framework



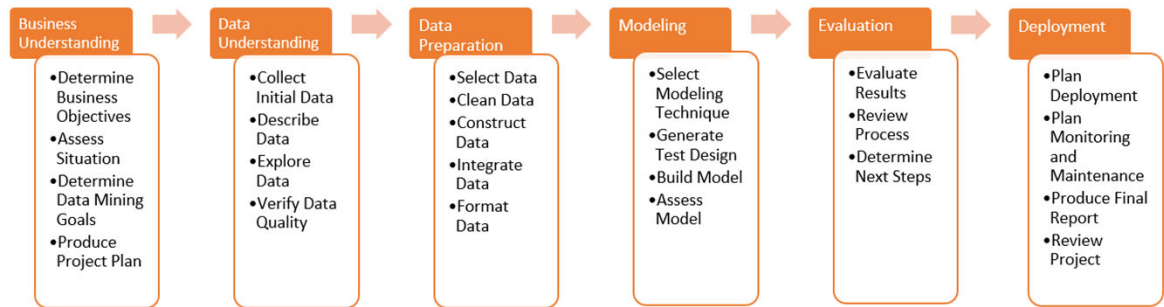
CRISP-DM

Cross Industry
Standard Process for
Data Mining

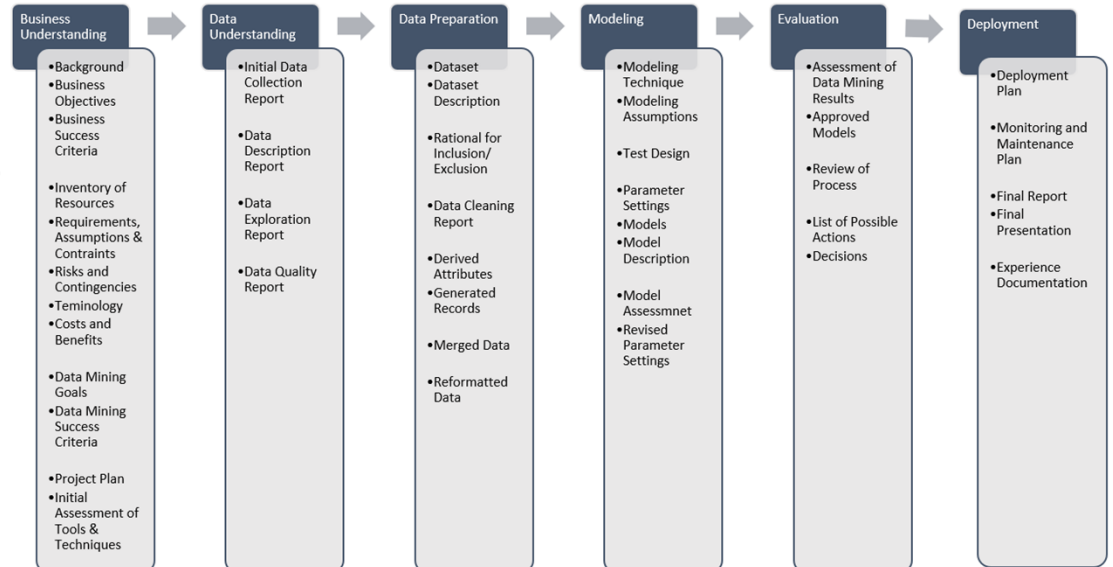


CRISP-DM: Tasks and Outputs

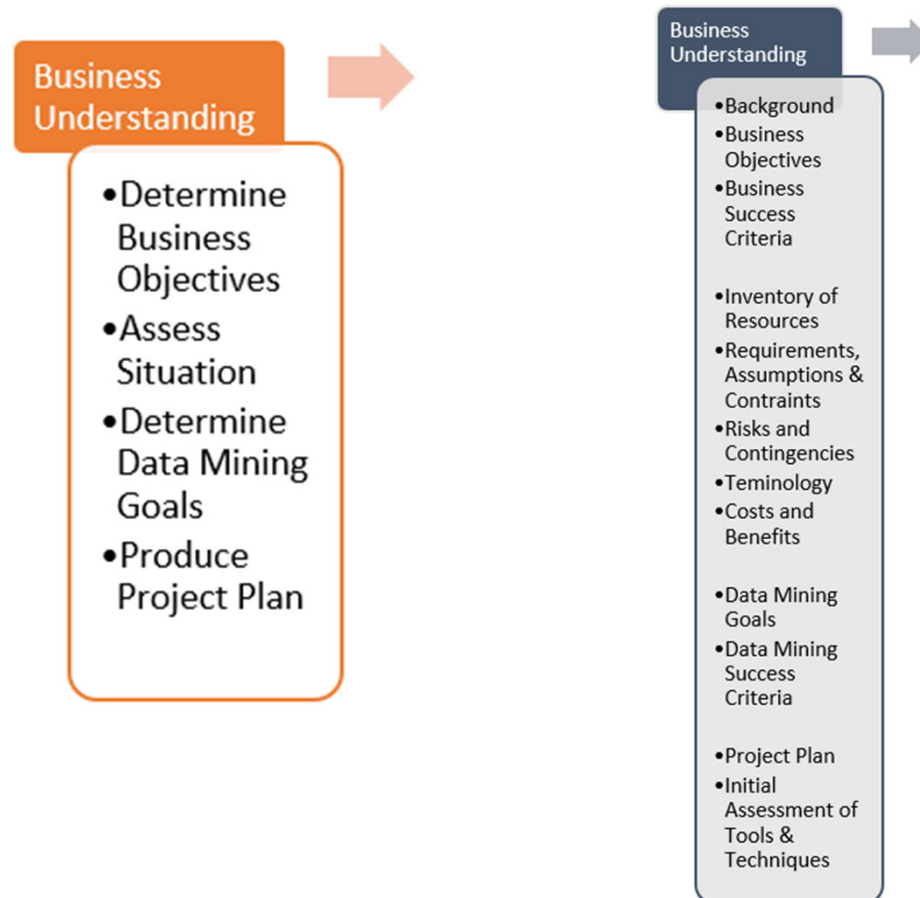
TASKS



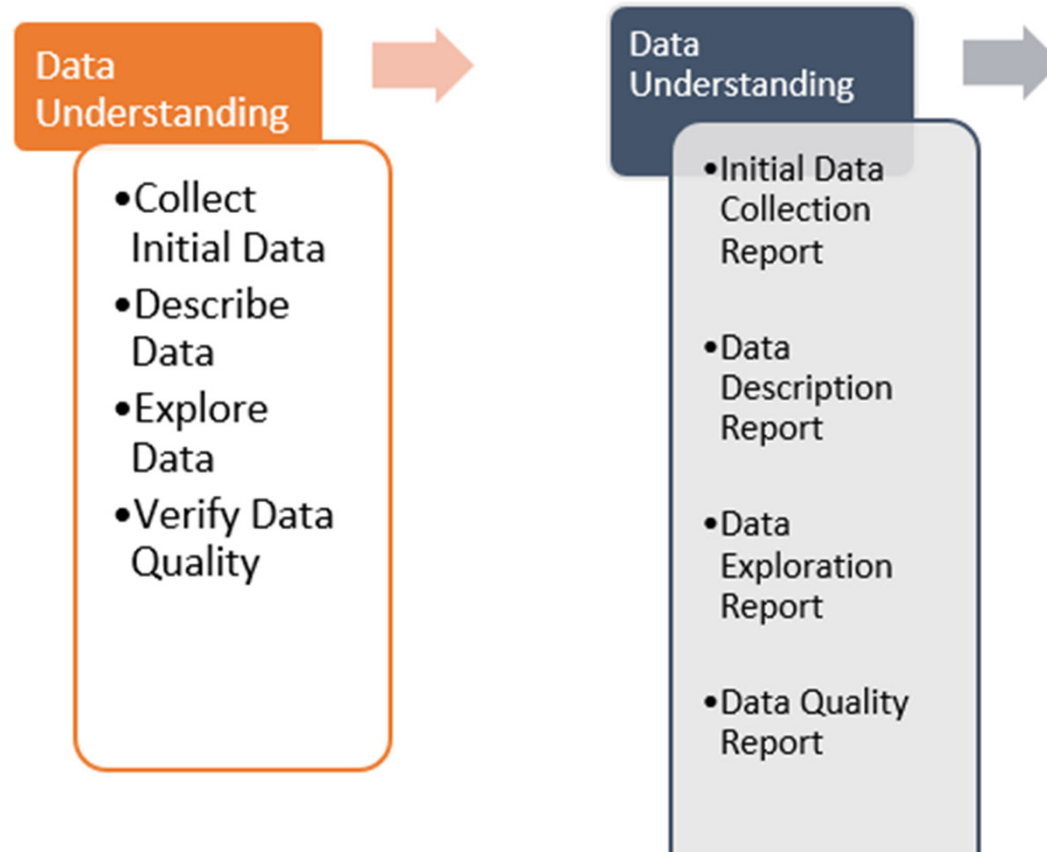
OUTPUTS



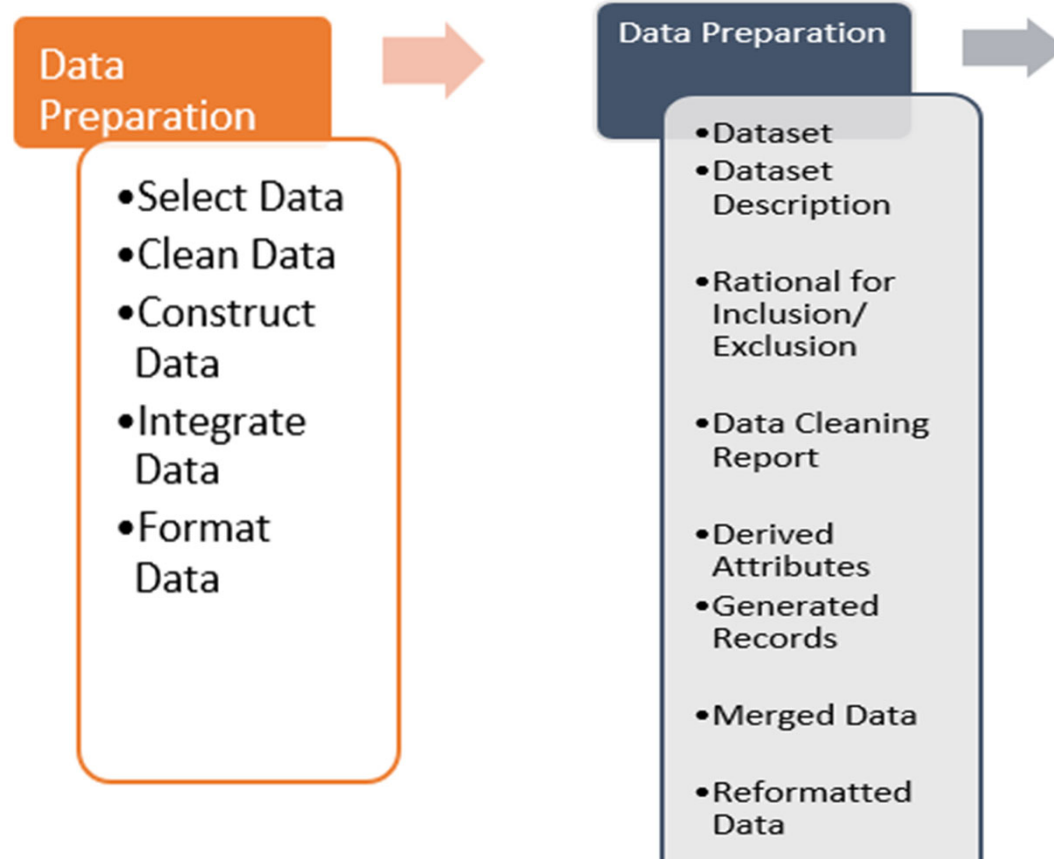
CRISP-DM: Business Understanding



CRISP-DM: Data Understanding



CRISP-DM: Data Preparation



Spreadsheet and Programming Tools for Business Analytics



Foundational Business Analytics Tools



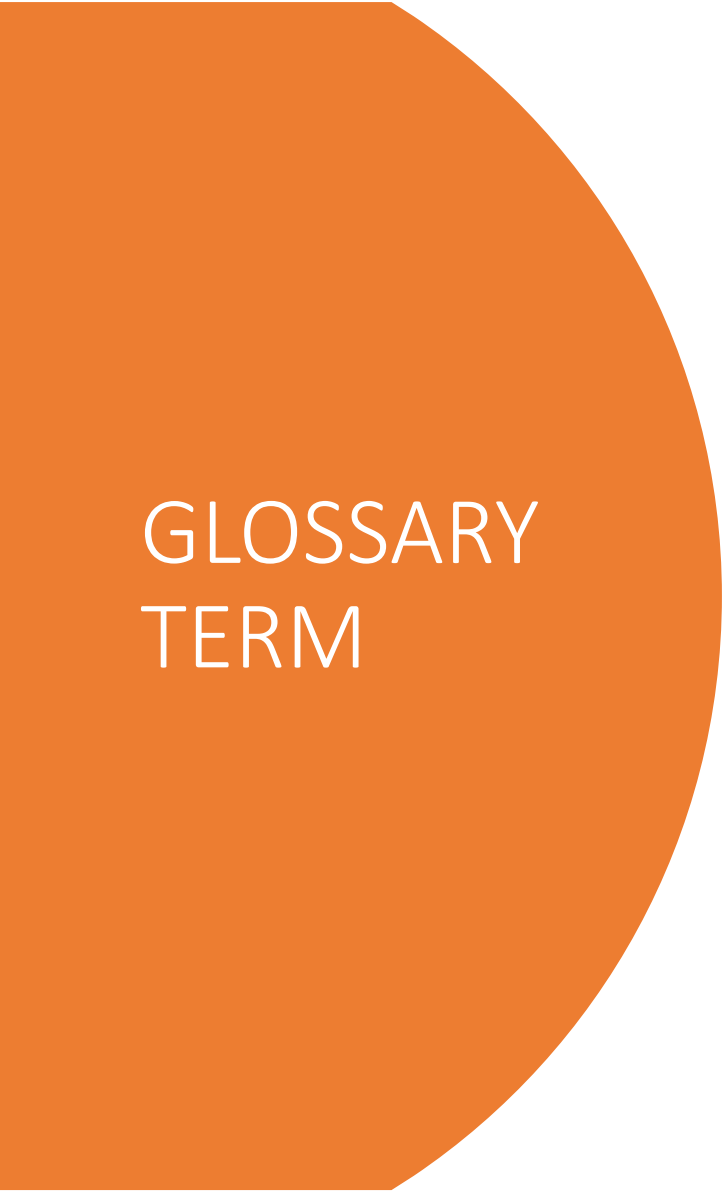
GENERAL SPREADSHEETS



PROGRAMMING TOOLS



PROPRIETARY BUSINESS
ANALYTICS SOLUTIONS

A large orange shape on the left side of the slide, consisting of a rectangle with a quarter-circle cutout on its right side.

GLOSSARY TERM

spreadsheet tool: an interactive software application for structuring, transforming, analyzing, and storing data in rows and columns



Opening Tabular Spreadsheet Data

- Data come in all shapes and sizes. XLS and XLSX are both native files that can be opened by Excel.
- A lot of data are already in tabular form. Tabular form means it has rows and columns.
- For example, in a Comma Separated Value file, each row of data is stored in a text file with a comma separating each column's values from one another.

Opening Tabular Spreadsheet Data (cont.)

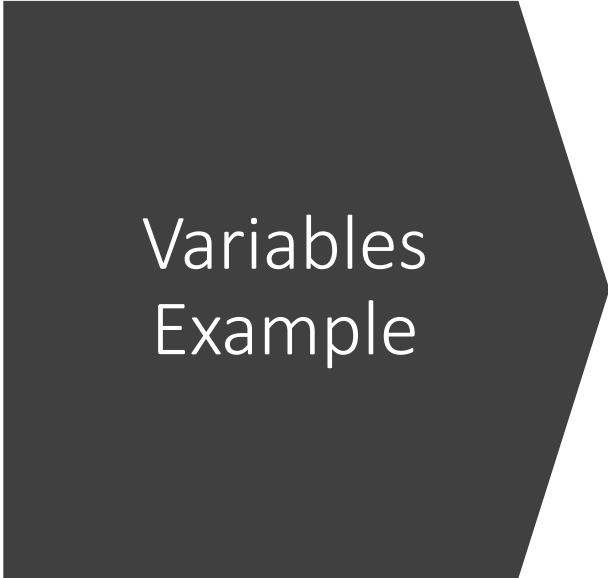
- Tab Delimited file: the columns of data are stored as a text file with a TAB character between values.
- TAB delimitation has an advantage over CSV: commas inside text values (e.g., “Smith, Rodrigo”) can confuse the software about where values actually begin and end.

Data Transformation Using Spreadsheets

Data can be cleaned by taking advantage of search and replace, sorting and filtering, and built-in functions.

5 Basic Tasks in Data Understanding & Cleaning

1. Load the data into a spreadsheet
2. Sort each column
3. Aggregate or create new columns
4. Think about whether a particular outcome makes sense
5. Make corrections directly in the spreadsheet



Variables Example

STUDENT	AGE	GPA	BUSINESS MAJOR
Jill	20	3.1	TRUE
Jerome	20	3.2	FALSE
Deepak	21	2.8	TRUE
Taylor	19	3.3	TRUE

GLOSSARY TERMS

programming: the process of solving a problem using computer algorithms

- **programming language:** a formal set of instructions that can be used to produce various kinds of output
- **open-source programming tools:** programming tools that are made freely available, often developed by and for the community
- **programming code:** a collection of statements written in a particular programming language